Isaac Marval Hernández

Software Development · Cybersecurity · Applied AI

Rivas Vaciamadrid, Madrid, Spain | +34 625 49 69 18 | iamarvalh@gmail.com | LinkedIn | GitHub | marvali.github.io

SUMMARY

Software engineer (Information Systems Engineering) with hands-on cybersecurity experience. Builds end-to-end applications in Python, Java and SQL, from REST APIs to applied AI, backed by cyber threat intelligence (OSINT, OPSEC) and sysadmin experience. Currently building a secure, on-premise LLM for confidential document analysis.

EDUCATION

BSc in Information Systems Engineering — Universidad de Alcalá (UAH) *(final year)* 2022 – 2027

Higher Technician in Telecom & Computer Systems — Salesianos de Atocha *(8.81/10, top of class)* 2020 – 2022

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, SQL, JavaScript, MATLAB, Bash, PowerShell, HTML, CSS
- **Software Engineering:** OOP, UML, REST APIs (FastAPI), full-stack web development, concurrent & distributed programming (Java threads, RMI), unit & black-box testing, Git, GitHub, Jira
- **Data:** PostgreSQL, relational data modeling, Power BI (data cleaning, dashboards), NumPy
- **AI & LLMs:** LLMs, Generative AI, Retrieval-Augmented Generation (RAG), on-premise LLM deployment, prompt engineering, machine learning (k-NN, SVM, neural networks, k-means), evolutionary algorithms, computer vision
- **AI-Assisted Development:** Claude Code (subagents, hooks, MCP, skills), GitHub Copilot, Codex
- **Security:** OSINT, cyber threat intelligence, OPSEC (sock puppets, Tails, compartmentalized identities), forensic acquisition (disk imaging, hash verification), network security fundamentals, Tor, Wireshark, Nmap (basic)
- **Systems & Networking:** Linux, Windows, Docker, virtualization, server administration, TCP/IP, routing & switching

WORK EXPERIENCE

Cyber Threat Intelligence Intern (OSINT) — Qintegra, Madrid Jan 2022 – Jun 2022

- OSINT monitoring of the technology sector and production of strategic intelligence reports.
- Created and maintained sock-puppet research personas under strict OPSEC: dedicated hardware, Tails (amnesic OS running from RAM), prepaid SIMs and segregated networks; trained in forensically sound data acquisition.

IT Technician (Internship) — OpenSpring S.L. / Datatronics, Madrid Jun 2022 – Sep 2022

- Deployed and administered VMs and servers (incl. VoIP) to client specs; corporate IT and network support.

SELECTED PROJECTS

Secure On-Premise LLM — Final Thesis (TFG) 2026 · in progress

Self-hosted LLM with hybrid RAG (BM25 + dense retrieval) to query confidential documents fully offline, with document-level access control, audit logging and prompt-injection / data-leakage testing, aligned with ENS / ISO 27001.

Local RAG for F1 Regulations (GitHub) Mar 2026

PDF ingestion, embeddings, FAISS vector search and local LLM inference in Python.

Recipe Social Network (GitHub) 2025/26

Full-stack web application: FastAPI REST API, PostgreSQL data model and HTML/CSS/JavaScript front end.

Additional projects: concurrent Java & RMI, computer vision, soft computing & optimization, OSINT — see marvali.github.io

CERTIFICATIONS & LANGUAGES

Training: Cisco NetAcad — CCNAv7 (Switching, Routing & Wireless Essentials), IT Essentials; Power BI course.

Languages: Spanish — native; English — C1.